

Radioactivity

Radioactivity:

Radioactivity was originally discovered by Becquerel in 1896. He discovered that uranium element -92 gave out some kind of radiations. This phenomenon is confined almost entirely to the heaviest element from 83 to 106 in the periodic table.

In 1898, it was found by French husband-and-wife team of Pierre Curie and Marie Curie that polonium and radium also emitted radioactive radiations.

The radioactive radiations emitted by these elements are found to consist of the following:

- i. α -particles
- ii. β -particles
- iii. γ - rays or photon.

This Phenomenon of spontaneous emission of powerful radiation exhibited by heavy elements is called radioactivity.

Radioactivity is essentially a nuclear phenomenon and is a drastic process because the elements changes in kind .It is spontaneous and an irreversible self – disintegration activity because the elements breaks it up for good.

Those elements which exhibit this activity are called radioactive elements. Examples are: uranium, polonium, radium, radon, ionium, thorium, actinium, and mesothorium.

Properties of Radioactive Radiations:

The Properties of Radioactive Radiations are given below:

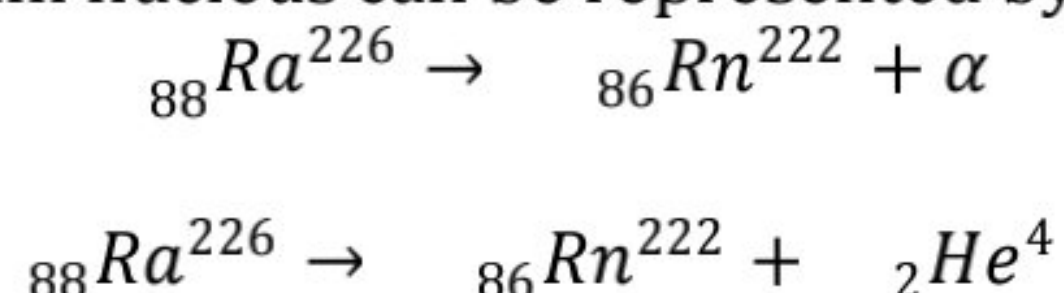
- i. These radiations are highly penetrating, could affect photographic plates.
- ii. As radiations are given out, new elements are formed in an irreversible process-the new elements themselves being usually radioactive.
- iii. The emission of radiations is spontaneous and is not affected by external agents.
- iv. The emission is not instantaneous but is prolonged i.e., it is extended over a period of time otherwise it would not have been discovered at all.
- v. Except for radioactivity, there is nothing abnormal about the radioactive elements as regards their physical and chemical properties.

Radioactive Disintegration or Decay:

Atoms of heavy elements like uranium, thorium, polonium and radium etc. are continuously breaking up into fresh radioactive atoms with the emission of α -, β - and γ -rays from their nuclei. In the process, the original (or parent) atom disappears and gives rise to new (daughter) atom. These new atoms are also, in general, radioactive and hence spontaneously break up, in their turn, thereby leading to a long chain of different radioactive elements in the form of a series until an inactive element (usually lead) is reached. This spontaneous breaking up of the nucleus is known as radioactive disintegration.

For example, when a radium (Ra) atom disintegrates by ejecting a α -particle, its nucleus loses positive charge of 2 units and mass of 4 units. Hence, it changes into a new atom called radon (Rn). It means that there occurs change of radium into radon.

The disintegration of radium nucleus can be represented by the following equation:-



The four radioactive series are given below:

Mass numbers	Series	Parent	Half-Life, years	Stable end Product
4n	Thorium	${}_{90}\text{Th}^{232}$	1.39×10^{10}	${}_{82}\text{Pb}^{208}$
4n+1	Neptunium	${}_{93}\text{Np}^{237}$	2.25×10^{10}	${}_{83}\text{Bi}^{209}$
4n+2	Uranium	${}_{92}\text{U}^{238}$	4.51×10^{10}	${}_{82}\text{Pb}^{206}$
4n+3	Actinium	${}_{92}\text{U}^{235}$	7.07×10^{10}	${}_{82}\text{Pb}^{207}$

Laws of Radioactive Disintegration:

According to Rutherford and Soddy theory, it is supposed that the radioactive atoms are unstable and they disintegrate according to the law of chance. The disintegration of a single atom is accompanied by the ejection of a single α - or β -particle.

Radioactive disintegration is found to obey the following laws:

- i. Atoms of all radioactive elements undergo spontaneous disintegration to form fresh radioactive products with the emission of α -, β - and γ -rays.
- ii. The rate of radioactive disintegrations i.e., the number of disintegrations per second is not affected by environmental factors (like temperature, pressure, and chemical combination etc.) but depends on the number of the atoms of the original kind present at any time.

Let us consider,

N_0 = The number of radioactive atoms present in a sample at the beginning of disintegration i.e., $t=0$.

N = The number of radioactive atoms present in a sample at any time t .

dN = The number of radioactive atoms present in a sample at a time dt .

At this instant, the rate of disintegration

$$\begin{aligned}\frac{dN}{dt} &\propto -N \\ \Rightarrow \frac{dN}{dt} &= -\lambda N \dots \dots \dots (1)\end{aligned}$$

Where λ is the constant of proportionality and is known as the disintegration constant. The negative sign shows that with the increase in disintegration the value of N decreases.

Rearranging the above equation, we get

$$\begin{aligned}\frac{dN}{N} &= -\lambda dt \\ \Rightarrow \int \frac{dN}{N} &= -\lambda \int dt \\ \Rightarrow \ln N &= -\lambda t + c\end{aligned}$$

where c is a constant.

When $t=0$, $N = N_0$

$$\therefore \ln N_0 = c$$

Substituting this value of c in equation (1)

$$\ln N = -\lambda t + \ln N_0$$

$$\Rightarrow \ln \frac{N}{N_0} = -\lambda t$$

$$\Rightarrow \frac{N}{N_0} = e^{-\lambda t}$$

$$\therefore N = N_0 e^{-\lambda t} \dots \dots \dots (2)$$

This equation represents the Laws of Radioactive Disintegration. **Thus, the number of radioactive atoms decreases exponentially with time** as shown in (Fig.-01).

Fig.-01

Half Life Period:

The half-life period of a radioactive substance is defined as the time required for one-half of the radioactive substance to disintegrate.

The half -life period is different for different substance and depends upon the radioactive constant of the substance.

We know,

$$N = N_0 e^{-\lambda t} \dots\dots\dots(1)$$

Here,

N_0 = The number of radioactive atoms present in a sample at the beginning of disintegration i.e., $t=0$.

N = The number of radioactive atoms present in a sample at any time t .

λ = Disintegration constant.

Suppose half-life period = $T_{1/2}$

Now, if $t = T_{1/2}$, then $N = \frac{N_0}{2}$

Therefore, from equation (1),

$$\frac{N_0}{2} = N_0 e^{-\lambda T_{1/2}}$$

$$\Rightarrow \frac{1}{2} = e^{-\lambda T_{1/2}}$$

$$\Rightarrow \ln\left(\frac{1}{2}\right) = -\lambda T_{1/2}$$

$$\Rightarrow \ln 1 - \ln 2 = -\lambda T_{1/2}$$

$$\Rightarrow \ln 2 = \lambda T_{1/2}$$

$$\Rightarrow T_{1/2} = \frac{\ln 2}{\lambda}$$

$$\therefore T_{1/2} = \frac{0.693}{\lambda} \dots\dots\dots(2)$$

The above equation represents the expression for half-life period.

As seen, $T_{1/2}$ is inversely proportional to the radioactive constant.

The Mean Life:

It is not possible to predict which atom of a radioactive substance will disintegrate at any instant. The atom which disintegrates first has zero life and that disintegrates last has infinite life. Thus the life of every atom is different and the actual lives of the various atoms range from zero to infinity.

The mean life of a radioactive element

$$= \frac{\text{Sum of the lives of all the atoms}}{\text{Total number of atoms}}.$$

Definition: The mean-life of a radioactive element defined as the ratio of the total life time of all the radioactive atoms to the total number of such atoms in it.

Value of mean-life:

Let N_0 be the total number of radioactive atoms in the beginning. Let N be the number of atoms of that element after time t . Then $N = N_0 e^{-\lambda t}$. Let dN be the number of atoms disintegrating between time t and $t + \delta t$. These dN atoms have had a life between t and $t + \delta t$. Since δt is very small, each of these atoms had a life of t .

$$\therefore \text{Total life of } dN \text{ atoms} = (dN)t$$

The possible life of any of the total number N_0 radioactive atoms varies from 0 to ∞ .

$$\therefore \text{Total life time of all } N_0 \text{ atoms} = \int_0^{\infty} t dN$$

$$\begin{aligned} \text{Now, mean life} = \bar{T} &= \frac{\text{total life-time}}{\text{total number of atoms}} \\ &= \frac{\int_0^{\infty} t dN}{N_0} \end{aligned}$$

Now,

$$\begin{aligned} N &= N_0 e^{-\lambda t} \\ \therefore \frac{dN}{dt} &= -\lambda N_0 e^{-\lambda t} \\ \Rightarrow dN &= -\lambda N_0 e^{-\lambda t} dt \\ \Rightarrow dN &= \lambda N_0 e^{-\lambda t} dt \end{aligned}$$

(Leaving the negative sign which merely indicates the decreases in the number of atoms with time).

$$\text{Hence, } \bar{T} = \frac{\int_0^{\infty} \lambda N_0 e^{-\lambda t} dt}{N_0} = \lambda \int_0^{\infty} t e^{-\lambda t} dt$$

Integrating by parts,

$$\begin{aligned} \bar{T} &= \left[\frac{t e^{-\lambda t}}{-\lambda} - \int \frac{e^{-\lambda t} dt}{-\lambda} \right] \\ \Rightarrow \bar{T} &= \lambda \left[\frac{t e^{-\lambda t}}{-\lambda} - \frac{e^{-\lambda t}}{-\lambda^2} \right] \end{aligned}$$

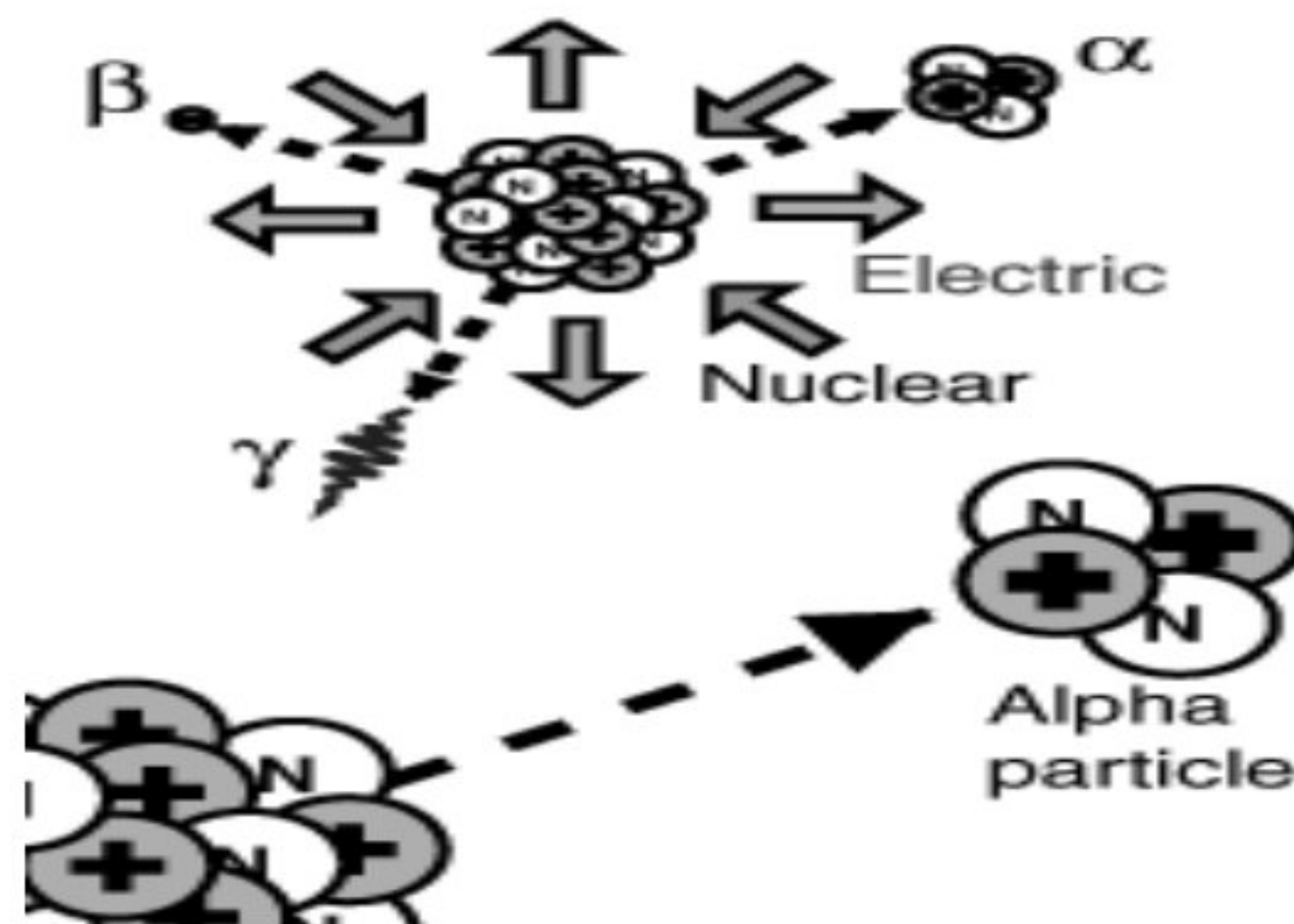
$$\begin{aligned} &= \frac{1}{\lambda} \\ \therefore \bar{T} &= \frac{1}{\lambda} \end{aligned}$$

Thus the mean life ($\bar{T}$) of a radioactive substance is the reciprocal of the decay constant (λ).

Alpha Radioactivity/ alpha particle/Alpha Rays/Alpha Decay:

A positively charged particle that consists of two protons and two neutrons bound together. It is emitted by an atomic nucleus undergoing radioactive decay and is identical to the nucleus of a helium atom. Because of their relatively large mass, alpha particles are the slowest and least penetrating forms of nuclear radiation. They can be stopped by a piece of paper.

It is not suitable for radiation therapy since its range is less than a tenth of a millimeter inside the body. Its main radiation hazard comes when it is ingested into the body; it has great destructive power within its short range. In contact with fast-growing membranes and living cells, it is positioned for maximum damage.

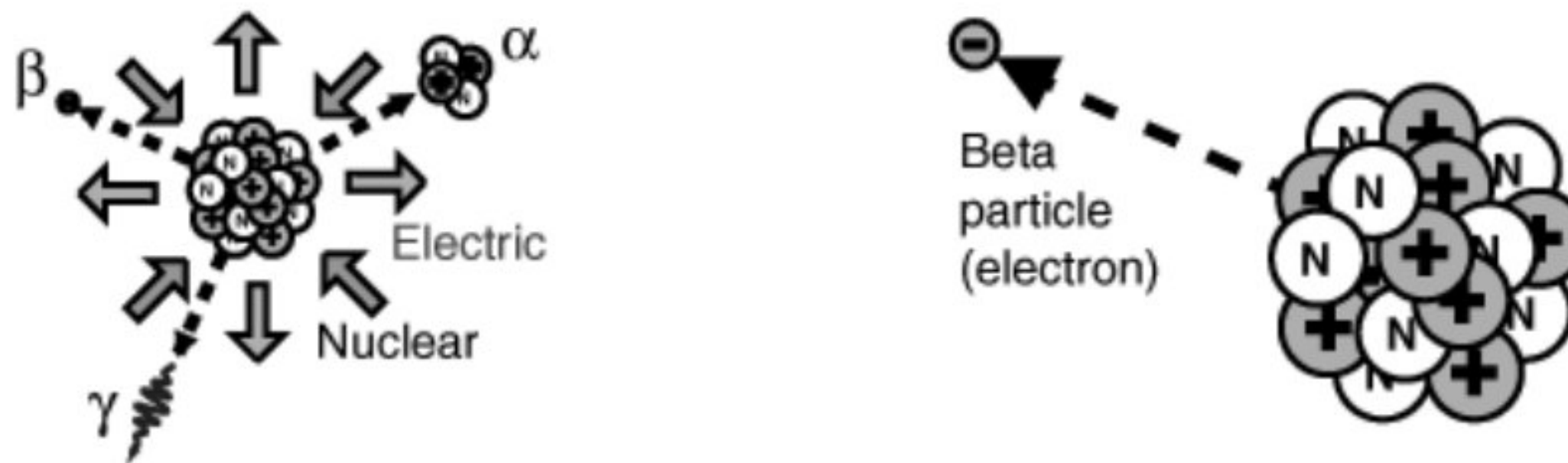


Alpha particle emission is modeled as a barrier penetration process. The alpha particle is the nucleus of the helium atom and is the nucleus of highest stability.

Beta Radioactivity/Beta Particles/Beta Rays/ Beta decay:

Beta particles are just electrons from the nucleus, the term "beta particle" being an historical term used in the early description of radioactivity. The high energy electrons have greater range of penetration than alpha particles, but still much less than gamma rays. The radiation hazard from betas is greatest if they are ingested.

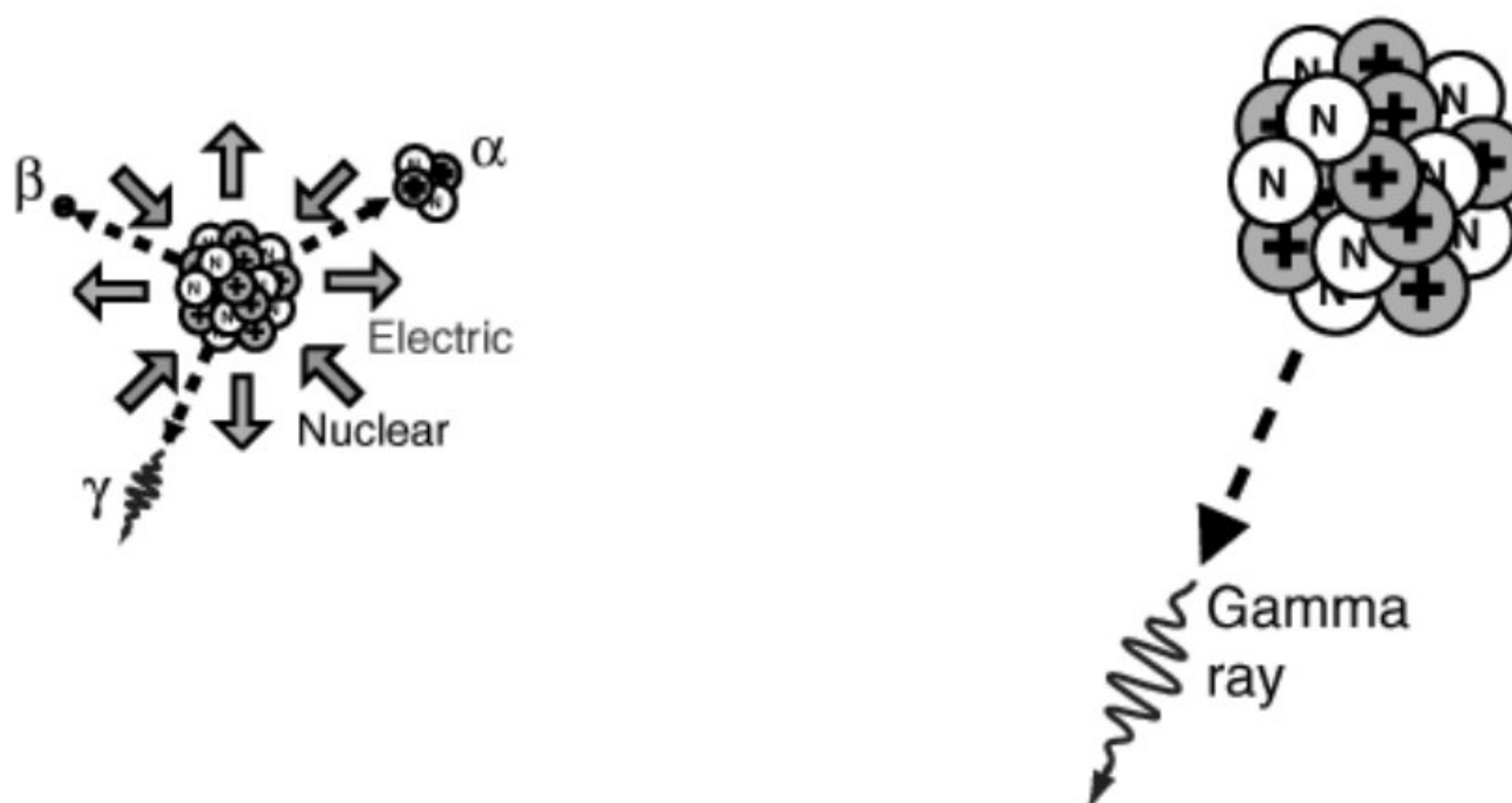
Beta emission is accompanied by the emission of an electron antineutrino which shares the momentum and energy of the decay.



Gamma Radioactivity/Gamma Particles/Gamma Rays/ Gamma decay:

A stream of high-energy electromagnetic radiation given off by an atomic nucleus undergoing radioactive decay. Because the wavelengths of gamma rays are shorter than those of x-rays, gamma rays have greater energy and penetrating power than x-rays. Gamma rays are emitted by pulsars, quasars, and radio galaxies but cannot penetrate the Earth's atmosphere.

It is the most useful type of radiation for medical purposes, but at the same time it is the most dangerous because of its ability to penetrate large thicknesses of material.

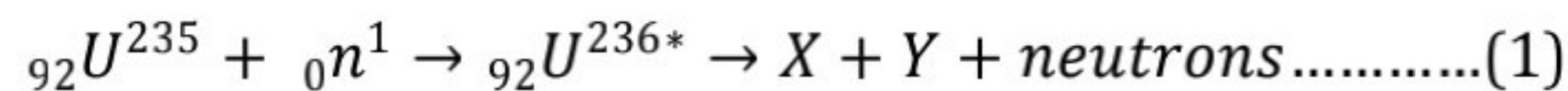


Nuclear Fission:

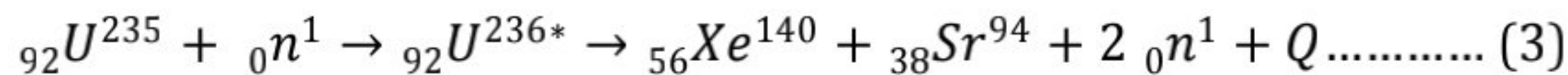
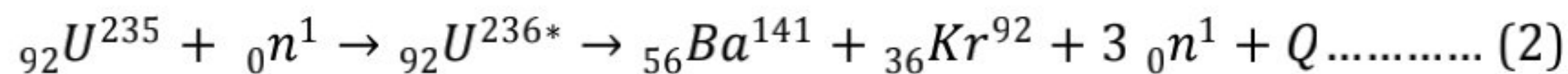
The process of breaking up of the nucleus of a heavy atom into two, more or less equal fragments with the release of a large amount of energy is known as fission.

When uranium is bombarded with neutrons, a uranium nucleus captures a slow neutron, forming an unstable compound nucleus. The compound nucleus splits into two nearly equal parts. Some neutrons are also released in this process.

The schematic equation for the fission process is



${}_{92}\text{U}^{236*}$ is a highly unstable isotope, and X and Y are the fission fragments. The fragments are not uniquely determined, because there are various combinations of fragments possible and a number of neutrons are given off. Typical fission reactions are



Where Q is the energy released in the reaction.

According to equation (2), when ${}_{92}\text{U}^{235}$ is bombarded by a slow moving neutron, the nucleus becomes unstable (${}_{92}\text{U}^{236*}$) and splits into ${}_{56}\text{Ba}^{141}$ and ${}_{36}\text{Kr}^{92}$ releasing 3 neutrons and energy Q(Fig.-01).

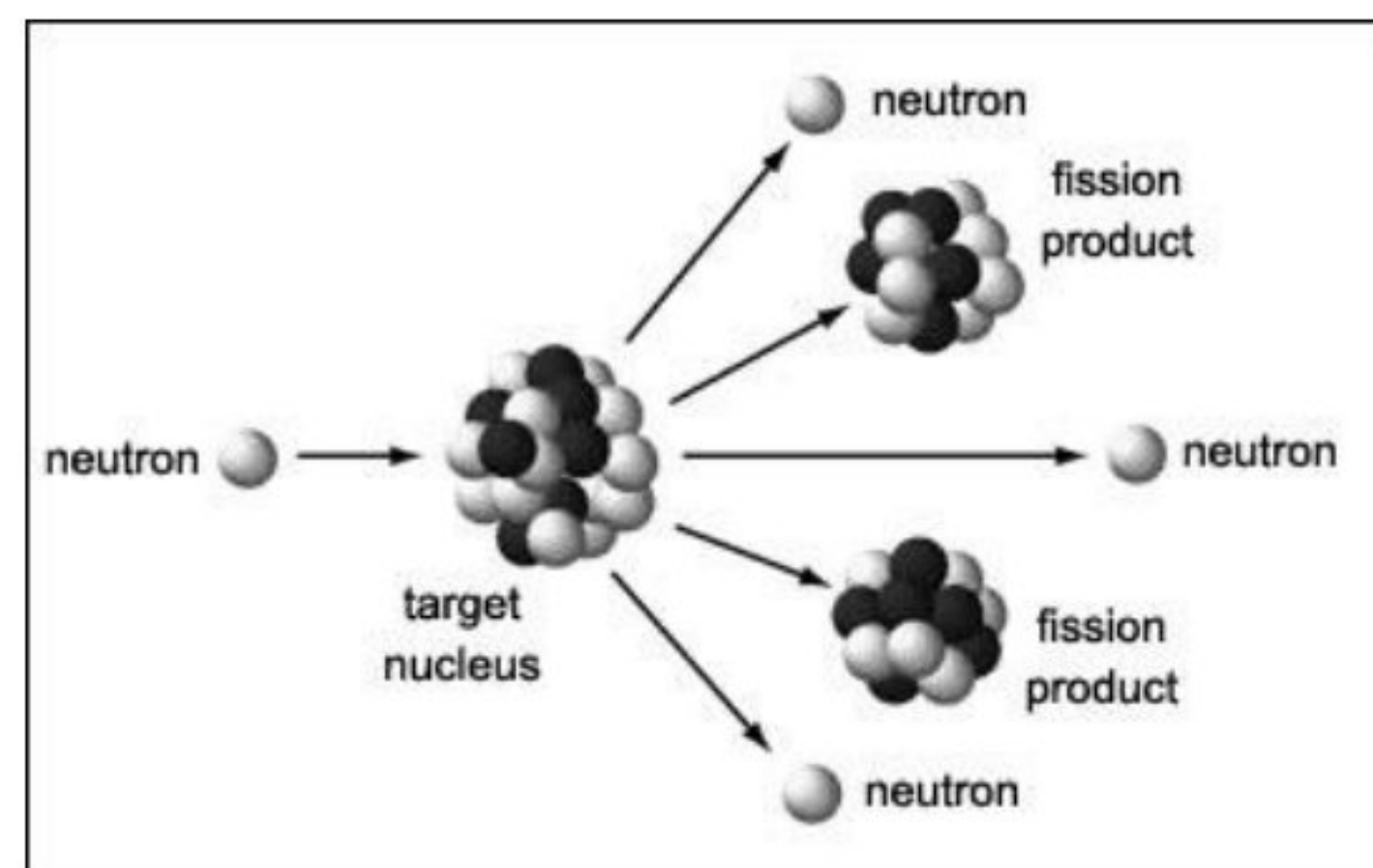


Fig.-01

Nuclear Fusion:

The process in which two or more light nuclei are combined into a single nucleus with the release of tremendous amount of energy is called as **nuclear fusion**. Like a fission reaction, the sum of masses before the fusion (i.e. of light nuclei) is more than the sum of masses after the fusion (i.e. of bigger nucleus) and this difference appears as the fusion energy. The most typical fusion reaction is the fusion of two deuterium nuclei into helium.



For the fusion reaction to occur, the light nuclei are brought closer to each other (with a distance of 10^{-14} m). This is possible only at very high temperature to counter the repulsive force between nuclei. Due to this reason, the fusion reaction is very difficult to perform. The inner core of sun is at very high temperature, and is suitable for fusion, in fact the source of sun's and other star's energy is the nuclear fusion reaction

Nuclear Cross Section:

A measure of the probability for a reaction to occur between a nucleus and a particle; it is an area such that the number of reactions which occur in a sample exposed to a beam of particles equals the product of the number of nuclei in the sample and the number of incident particles which would pass through this area if their velocities were perpendicular to it.

